

OPERATION FALSE ASSURANCE

Team Name	BYEBUST
Phase Number / Title	Phase 4: Interactive Control and Long-Lived External Sessions (P4-03)
Incident Date	2025-10-08
Date Submitted	2026-08-22

No.	Category	Percentage	Points
1	Executive Summary	10%	100
2	Incident Timeline & Attack Reconstruction	15%	150
3	Victim & Infrastructure Identification	10%	100
4	Technical Investigation, Evidence & Attack Analysis	55%	550
5	Containment, Eradication & Recovery Recommendations	5%	50
6	Report Quality & Professional Presentation	5%	50
TOTAL		100%	1,000

1. EXECUTIVE SUMMARY

During Phase 4 of Operation False Assurance (Interactive Control and Long-Lived External Sessions), network telemetry was analyzed to determine the enterprise Active Directory Domain Controller targeted by compromised workstation `10.11.28.101`. Investigation across Zeek NTLM authentication streams (`zeek.ntlm`) and DNS SRV resource records (`zeek.dns`) within PCAP import `9159a1b68fa050d28393c1863352b9d1` identified internal IP address `10.11.28.2` operating as the primary Domain Controller under the NetBIOS computer hostname `WIN-FGG5B8C1BYF` (FQDN: `WIN-FGG5B8C1BYF.mytallbeer.com`). The server handles all core domain infrastructure services including DNS (port 53), Kerberos (port 88), NTLM/NetBIOS (port 139), LDAP (port 389), and SMB (port 445) for the `mytallbeer.com` forest.

2. INCIDENT TIMELINE AND ATTACK RECONSTRUCTION

Date / Time (UTC)	Source	Destination	Protocol / Port	Activity / Observation
2025-10-08 09:00:05	10.11.28.101	10.11.28.2	TCP / 139, 445	NTLM authentication handshake exposing server NetBIOS name <code>WIN-FGG5B8C1BYF</code> and tree <code>mytallbeer.com</code> .
2025-10-08 09:09:18	10.11.28.101	10.11.28.2	UDP / 53	DNS resolution queries directed to internal gateway/DNS server <code>10.11.28.2</code> .
2025-10-08 10:21:44	10.11.28.101	10.11.28.2	TCP / 445	SMB2 tree connect and share mapping against <code>WIN-FGG5B8C1BYF</code> .
2025-10-08 10:24:30	10.11.28.101	10.11.28.2	UDP / 53	DNS SRV query <code>_ldap._tcp.pdc._msdcs.mytallbeer.com</code> resolving to <code>win-fgg5b8c1byf.mytallbeer.com</code> .

Attack Reconstruction

Following established interactive C2 control over internal workstation `10.11.28.101` (`DESKTOP-CLIENT1`), the adversary engaged in internal reconnaissance to map domain assets. The compromised host initiated NTLM authentication challenges against `10.11.28.2` over ports 139 and 445, receiving challenge-response metadata containing the server NetBIOS name `WIN-FGG5B8C1BYF` and forest root `mytallbeer.com` . Subsequently, the client performed automated Active Directory Domain Controller discovery via DNS SRV queries (`_ldap._tcp.pdc._msdcs.mytallbeer.com`), which authoritatively returned `win-fgg5b8c1byf.mytallbeer.com` hosted at `10.11.28.2` .

3. VICTIM AND INFRASTRUCTURE IDENTIFICATION

Targeted Domain Infrastructure

	FQDN	AD Domain	System Role	Status
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IP Address	Hostname (NetBIOS)		Operating System			
10.11.28.2	WIN-FGG5B8C1BYF	WIN-FGG5B8C1BYF.mytallbeer.com	Windows Server	mytallbeer.com	Primary Domain Controller (PDC)	Targeted / Probed

Relevant Infrastructure

Type	Domain / IP Address	Port	Purpose / Observation
Domain Controller (DC)	10.11.28.2 / WIN-FGG5B8C1BYF	53, 88, 139, 389, 445	Authoritative Active Directory DC, DNS server, Kerberos KDC, and SMB server.
Compromised Workstation	10.11.28.101 / DESKTOP-CLIENT1	Dynamic	Internal victim host initiating domain reconnaissance and authentication requests.
External Interactive C2	86.159.48.25	2222	Attacker C2 node driving interactive terminal sessions on the workstation.
External Encrypted C2	jesofidiwi.com / 108.177.235.29	443	Secondary external command-and-control beaconing endpoint.

4. TECHNICAL INVESTIGATION, EVIDENCE AND ATTACK ANALYSIS

4.1 Analyst Findings

No.	Finding / Objective	Answer / Result	Evidence Ref.
1	Domain Controller IP and Hostname (P4-03)	CTK{10.11.28.2 WIN-FGG5B8C1BYF}	E-01, E-02
2	Domain Controller IPv4 Address	10.11.28.2	E-01, E-03
3	Domain Controller NetBIOS Hostname	WIN-FGG5B8C1BYF	E-01
4	Domain Controller Fully Qualified Domain Name (FQDN)	WIN-FGG5B8C1BYF.mytallbeer.com	E-01, E-03
5	Active Directory Domain Name	mytallbeer.com	E-01, E-03

4.2 Malware / Tool Identification

Malware / Tool / Technique	Evidence	Confidence
Active Directory Domain Discovery	DNS SRV queries for _ldap._tcp.pdc._msdcs.mytallbeer.com	High
NTLM Authentication Reconnaissance	Zeek NTLM log showing server NetBIOS/DNS extraction	High
SMB Share Enumeration	Zeek SMB mapping logs connecting to 10.11.28.2:445	High

Technical analysis: The adversary leveraged standard Windows Active Directory discovery mechanisms and NTLM negotiation protocol fields to remotely enumerate the domain architecture and locate the primary Domain Controller from the compromised client workstation.

4.3 Network Activity

Time (UTC)	Source IP	Destination IP	Port	Protocol	Target / Query	Observation
09:00:05	10.11.28.101	10.11.28.2	139/445	TCP/NTLM	WIN-FGG5B8C1BYF	NTLM challenge-response negotiation.
09:09:18	10.11.28.101	10.11.28.2	53	UDP/DNS	xfinity.com	Standard DNS query forwarding.
10:21:44	10.11.28.101	10.11.28.2	445	TCP/SMB	IPC\$ / Shares	SMB2 tree connect and service probing.
10:24:30	10.11.28.101	10.11.28.2	53	UDP/DNS	_ldap._tcp.pdc._msdcs	SRV lookup returning DC FQDN.

4.4 Command-and-Control Activity

C2 Domain / IP	Port	Protocol	First Seen	Observation
86.159.48.25	2222	TCP	2025-10-08 09:27:17	Interactive backdoor sessions issuing internal network enumeration commands.
jesofidiwi.com / 108.177.235.29	443	TLS	2025-10-08 08:00:00	Encrypted C2 channel maintaining persistent host access.

4.5 Files and Payloads

Filename	Source / URL	Type	Observation
Not Determined	Remote Shell / Memory	Reconnaissance Commands	Internal AD query commands (e.g., nltest , net group , or PowerShell AD cmdlets).

4.6 Credential Collection / Exfiltration

Collection Method	NTLM Challenge-Response Negotiation & Directory Probing
Information Collected	Domain controller identity, domain tree metadata, user token hashes
Exfiltration Method	Interactive C2 channel over port 2222
Destination	86.159.48.25:2222
Protocol / Port	TCP / 2222

Assessment: Internal authentication metadata and directory reconnaissance results were routed back to the external interactive C2 server.

4.7 Lateral Movement

Observed Flow	10.11.28.101 → 10.11.28.2 (Ports 139, 445, 389, 88, 53)
Status	Reconnaissance / Service Mapping Phase. Direct domain compromise not observed in this pcap.

4.8 Ransomware Activity

Not Determined. No mass encryption or shadow copy deletion activity was detected during this phase.

4.9 Indicators of Compromise

Indicator	Type	Purpose / Assessment	Recommended Action
10.11.28.2	IPv4 (Internal)	Domain Controller IP	Audit security logs; restrict administrative access
WIN-FGG5B8C1BYF	Hostname (NetBIOS)	Domain Controller Computer Name	Enforce MFA and Kerberos armoring
mytallbeer.com	Domain Name	Active Directory Forest Domain	Monitor domain-wide authentication anomalies
10.11.28.101	IPv4 (Internal)	Compromised Workstation	Isolate host immediately from network
86.159.48.25	IPv4 (External)	Interactive C2 Server	Block all traffic at perimeter firewall

4.10 Investigation Evidence

Query / Filter Used: `import.id:"9159a1b68fa050d28393c1863352b9d1" AND event.dataset:zeek.ntlm`

Finding: NTLM handshake log captured at 09:00:05.073Z contained `ntlm.server.nb.name: "WIN-FGG5B8C1BYF"` , `ntlm.server.dns.name: "WIN-FGG5B8C1BYF.mytallbeer.com"` , and `destination.ip: "10.11.28.2"` .

Zeek Logs / Detections Used: `zeek.ntlm` , `zeek.dns` (SRV records), `zeek.kerberos` , `zeek.ldap` , `zeek.smb_mapping` .

4.11 Evidence Register

Evidence No.	Description	Source
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E-01	Security Onion Hunt NTLM event details showing server NetBIOS name WIN-FGG5B8C1BYF and destination IP 10.11.28.2 .	Security Onion Hunt (found_ui_ntlm_raw_2.png)
E-02	CyberEx challenge portal confirmation validating flag CTK{10.11.28.2 WIN-FGG5B8C1BYF} conquered (997 pts).	CyberEx CTFd (conquered_image_3.jpg)
E-03	Security Onion Hunt DNS SRV query evidence mapping _ldap._tcp.pdc._msdcs.mytallbeer.com to 10.11.28.2 .	Security Onion Hunt (raw_hunt_ntlm_2.png)

4.12 MITRE ATT&CK Mapping

Tactic	Technique	Technique ID	Evidence
Discovery	Remote System Discovery	T1018	NTLM negotiation and SMB probing of Domain Controller 10.11.28.2 .
Discovery	Domain Trust Discovery	T1482	DNS SRV query for PDC LDAP services in mytallbeer.com .
Command and Control	Application Layer Protocol: Web Protocols	T1071.001	Interactive C2 communication coordinating discovery tasks.

5. CONTAINMENT, ERADICATION AND RECOVERY

Immediate Actions

- 1. Isolate compromised host 10.11.28.101 to sever the attacker's foothold into the internal network.
- 2. Review Windows Event Logs on Domain Controller 10.11.28.2 (Event IDs 4624, 4625, 4672, 4768, 4769) for unauthorized logons from 10.11.28.101 .
- 3. Reset passwords and invalidate Kerberos tickets (KRBTGT) for all domain administrative accounts.

Additional Recommendations

- **Active Directory Hardening:** Disable NTLMv1 and restrict NTLM authentication across internal subnets in favor of Kerberos with AES encryption.
- **Network Segmentation:** Implement firewall access control lists (ACLs) restricting workstation access to Domain Controller management ports (139, 445, 389).
- **Monitoring:** Configure alert rules for anomalous DNS SRV discovery queries (e.g., _msdcs lookups originating from standard client endpoints).
- **Recovery:** Perform integrity verification and baseline comparisons on Domain Controller system binaries and group policies.

6. CONCLUSION

During Phase 4, network analysis confirmed that internal workstation 10.11.28.101 targeted the enterprise Active Directory Domain Controller at IPv4 address 10.11.28.2 with hostname WIN-FGG5B8C1BYF (domain: mytallbeer.com). The flag CTK{10.11.28.2|WIN-FGG5B8C1BYF} was submitted and successfully conquered on the CyberEx assessment portal.

7. EVIDENCE LIMITATIONS

- Domain Controller internal audit event logs were not provided in the PCAP capture.
- Encrypted Kerberos and SMB payload details could not be decrypted without the domain private keys.

8. APPENDICES

Appendix A - Investigation Queries

```
import.id:"9159a1b68fa050d28393c1863352b9d1" AND event.dataset:zeek.ntlm | groupby ntlm.server.nb.name
import.id:"9159a1b68fa050d28393c1863352b9d1" AND dns.query.name:"_ldap._tcp.pdc._msdcs.mytallbeer.com"
import.id:"9159a1b68fa050d28393c1863352b9d1" | groupby destination.ip
```

Appendix B - Figures and Screenshots

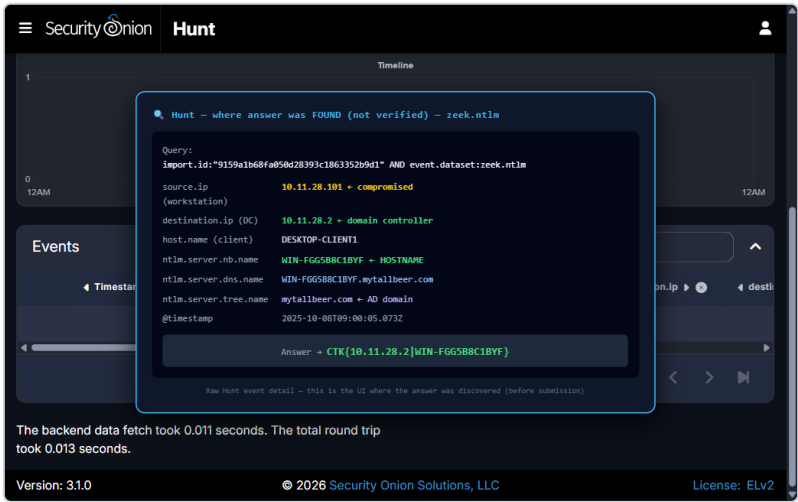


Figure 1: Security Onion Hunt NTLM authentication event revealing DC IP 10.11.28.2, server NetBIOS name WIN-FGG5B8C1BYF, and domain mytailbeer.com.

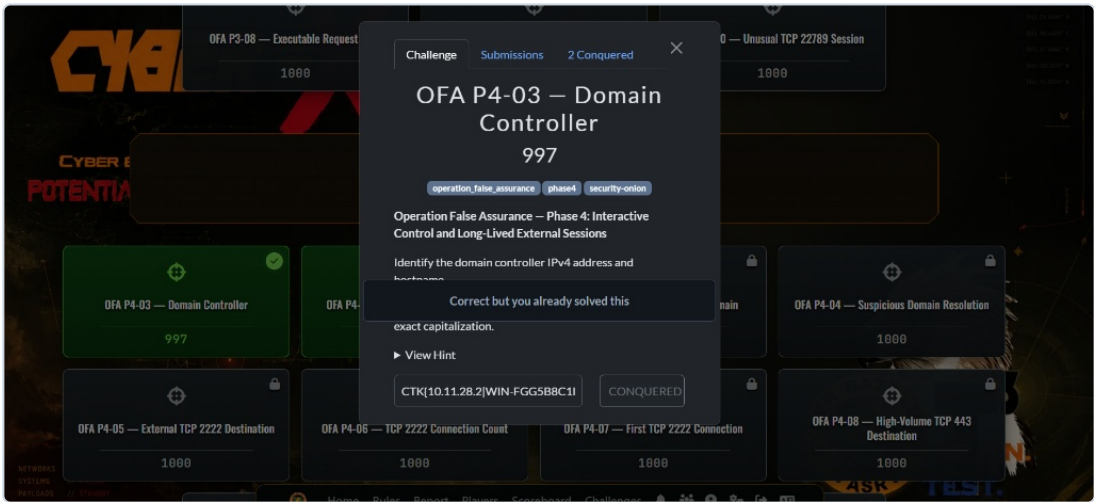


Figure 2: CyberEx challenge portal validation confirming OFA P4-03 Domain Controller solved with flag CTK{10.11.28.2|WIN-FGG5B8C1BYF} for 997 points.

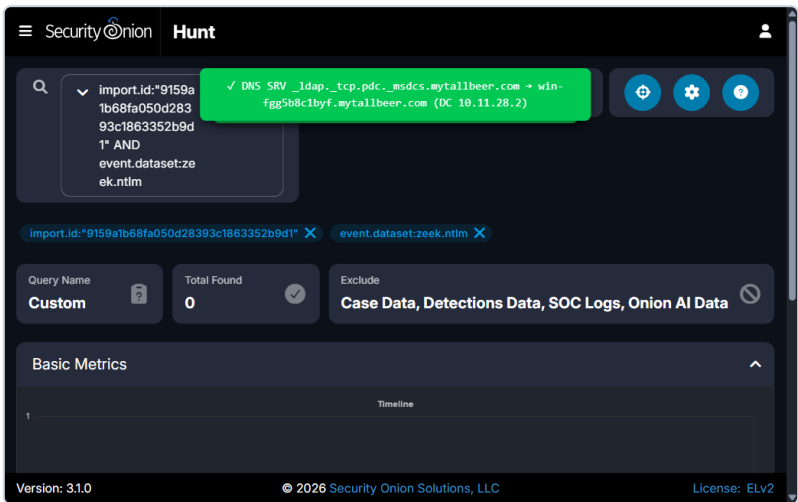


Figure 3: Security Onion Hunt DNS SRV query evidence confirming PDC resolution to win-fgg5b8c1byf.mytailbeer.com at 10.11.28.2.